ChronosLOB: Leakage-Safe Limit Order Book Forecasting with Execution-Aware Diagnostics

Anannye Naik

Abstract

Limit order book forecasting is often summarised by predictive metrics, but short-horizon accuracy is only one component of market-microstructure signal quality. A forecast with strong macro-F1 or MCC can change interpretation once probability calibration, selective prediction, active coverage, latency and execution frictions are brought into the evaluation. This paper presents ChronosLOB, a leakage-safe research platform that studies this forecasting-versus-signal-quality gap as a first-order empirical object. On FI-2010, a completed 180-cell validation-selected neural benchmark shows the matrix transformer outperforming a DeepLOB-style CNN–LSTM reference across 90 runs per family, with mean macro-F1 0.6013 ± 0.1950 versus 0.5133 ± 0.0670 and mean MCC 0.4294 ± 0.2727 versus 0.3356 ± 0.0883. A second-generation market-state SSL objective adds positive matched deltas across 30 comparison cells: macro-F1 $+0.0112$, MCC $+0.0259$, ECE -0.0030 and Brier score -0.0235. The central contribution is a rigorous validation framework for market-microstructure learning: predictive metrics are reported together with temporal leakage controls, calibration, feature stability, replay diagnostics and execution-aware stress tests.

1. Introduction

Market microstructure prediction sits at the intersection of noisy sequence modelling, non-stationary time series, strict temporal ordering and adversarially sparse signal. A limit order book evolves through liquidity provision, cancellation, queue depletion and trade arrival; a forecasting model observes only a finite state history and must infer short-horizon price movement from a process dominated by stochastic order flow. In this setting, model quality is not determined by a single score. It depends on how the data are constructed, how future labels are separated from past features, how probabilities are calibrated, how selective predictions behave and how quickly apparent signal decays under costs and latency.

ChronosLOB is built around a demanding research thesis: *forecasting quality and signal quality are distinct empirical objects*. A classifier can improve macro-F1 while becoming less useful under confidence filtering. It can produce attractive held-out metrics while concentrating its high-confidence predictions on a vanishingly small active set. It can exploit a powerful snapshot-derived proxy without measuring true event-level order flow. A serious market-microstructure paper should make these distinctions quantitative rather than treating them as caveats after the fact.

The platform is organised as an end-to-end empirical system. FI-2010 supplies the public benchmark backbone. Classical models, matrix transformers and DeepLOB-style networks provide supervised references. First-generation self-supervised objectives establish a null result for generic reconstruction-style pretraining; SSL-v2 then tests a more market-aware multitask objective. Feature ablations measure the stability of predictive information under controlled removals. Synthetic event replay validates event-level feature and label machinery under known regimes. Aggregate L2 replay validates local-book reconstruction mechanics. Execution-aware diagnostics stress forecasts through confidence thresholds, active fraction, turnover proxies, configured costs, row-step latency and adverse-selection proxies.

The paper's organising principle is precision without defensiveness. Each empirical statement is tied to its dataset, fold, horizon, seed, lookback, model family and diagnostic layer. Strong results can therefore be presented directly, while heterogeneous results remain scientifically interpretable. This gives ChronosLOB the character of a research instrument rather than a collection of experiments.

Contributions. This paper makes five contributions.

1. A leakage-safe FI-2010 pipeline with chronological folds, future-horizon labels, train-only preprocessing and matched-comparison discipline.
2. A completed 180-cell validation-selected neural benchmark comparing matrix transformer and DeepLOB-style models across folds, seeds, lookbacks and horizons, including dispersion across the full grid.
3. A market-state SSL-v2 objective evaluated across 30

matched FI-2010 comparison cells, with predictive and probabilistic deltas reported jointly.

4. Feature-stability, synthetic event-level replay and aggregate L2 reconstruction studies that connect static benchmark forecasting with broader market-data infrastructure.

5. An execution-aware diagnostic suite showing that confidence, activity, cost and latency materially change the interpretation of the same predictive forecasts.

2. Related Work

2.1. Limit Order Books and Short-Horizon Forecasting

A limit order book records standing bid and ask interest across price levels. Its state encodes spread, depth, imbalance and liquidity concentration, while its evolution reflects order submission, cancellation and execution dynamics. Short-horizon mid-price forecasting from LOB states is a standard empirical task because the target is mechanically defined, highly local and sensitive to microstructure state. FI-2010 is widely used for reproducible comparison because it provides processed LOB snapshots and fold conventions for mid-price forecasting (Ntakaris et al., 2018). Its value lies in reproducibility: it enables controlled comparisons under a public benchmark and removes many acquisition barriers that affect order-book research.

The same properties that make FI-2010 useful also define its scope. It is a snapshot benchmark, not a full event log. It can support leakage-safe forecasting, calibration analysis and snapshot-derived feature studies; it cannot directly reconstruct individual queues, order identities or realised fills. CHRONOSLOB handles this by separating snapshot benchmark results from replay and diagnostic studies. The platform does not collapse these layers into a single number.

2.2. Deep Architectures for LOB Data

DeepLOB-style models combine convolutions across price-level structure with temporal modules over recent snapshots (Zhang et al., 2019). This architecture is a natural reference point because LOB matrices contain both cross-sectional structure and time dependence. Transformers provide an alternative sequence-learning paradigm through attention (Vaswani et al., 2017), and are especially attractive when model interpretation depends on long-range context, multi-horizon targets or representation reuse. Sirignano and Cont (Sirignano & Cont, 2019) further show how deep learning can reveal recurring structure in price formation across markets.

In market data, architectural sophistication is insufficient unless it is paired with disciplined evaluation. Financial time series are susceptible to label overlap, leakage through preprocessing, unstable class balance and horizon-specific effects. CHRONOSLOB therefore treats model class as one component of a larger empirical protocol. The matrix transformer and DeepLOB-style family are compared under a validation-selected benchmark with matched folds, seeds, horizons and lookbacks.

2.3. Order Flow, Imbalance and Calibration

Order flow and imbalance are central to market microstructure. Cont, Kukanov and Stoikov (Cont et al., 2014) show that order-flow imbalance has substantial explanatory power for short-interval price changes when event-level order book data are available. Gould et al. (Gould et al., 2013) survey the broader structure of LOB dynamics, including spread, depth, queueing and liquidity provision. These ideas motivate features such as spread, depth imbalance, microprice, liquidity concentration and order-flow proxies.

Calibration is equally important. A model that reports probabilities should be judged not only by class decisions but also by whether its probabilities correspond to empirical frequencies. Dawid (Dawid, 1982), Brier (Brier, 1950) and Guo et al. (Guo et al., 2017) establish the statistical foundation and modern neural-network relevance of calibrated probability forecasts. In financial ML, chronological and purged evaluation principles help control leakage when labels overlap across time (Lopez de Prado, 2018). CHRONOSLOB combines these ideas into one validation stack: leakage-safe features and labels, predictive metrics, calibration, selective prediction and execution-aware stress tests.

3. System Overview

CHRONOSLOB is organised around the architecture in Figure 1. The data layer supplies three distinct data and diagnostic sources: FI-2010 snapshots, controlled synthetic event streams and aggregate L2 snapshot-plus-diff records. The feature and label layer enforces temporal availability: features are computed from information available at or before the prediction index, while labels use future horizons. The model layer contains both classical and neural families. The forecast layer records predictive and probabilistic metrics. The diagnostic layer tests how predictions behave under confidence and simplified execution frictions. The results layer records provenance, report freshness and scope.

This structure is intentionally modular. A benchmark result can be reproduced without invoking replay extensions. A replay report can validate book reconstruction without changing FI-2010 forecasting claims. Execution diagnostics can be regenerated from summaries. This modularity is valuable because market-microstructure research often mixes different result types. CHRONOSLOB keeps them aligned while preventing accidental inflation of what a given

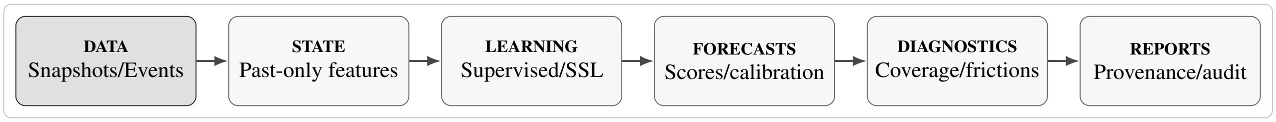

Figure 1. CHRONOSLOB validation architecture. The pipeline makes the temporal role of each stage explicit, from past-only state construction through probabilistic forecasting, selective diagnostics and reproducible reporting.

report can support.

3.1. Data and Diagnostic Sources

Table 1 summarises the data and diagnostic sources used in the paper. FI-2010 provides the main public forecasting benchmark. The synthetic extension provides controlled event-level dynamics where event OFI, cancellation imbalance, trade imbalance and regime labels are known. The aggregate L2 extension validates snapshot-plus-diff reconstruction mechanics. The execution diagnostic suite integrates forecasting outputs with confidence and friction diagnostics. The provenance reports record how the tables and figures were generated.

3.2. Provenance Layer

The reporting layer records how each major table and figure was generated. Benchmark summaries, configuration snapshots, command entry points and validation reports form a provenance trail from manuscript statements to executable code. This design is practical rather than ceremonial: market-data experiments often produce large prediction files and checkpoints, while public repositories usually expose compact summaries. CHRONOSLOB keeps the public paper inspectable by linking results to reports and manifests rather than relying on opaque prose.

The same layer also separates complete benchmark outputs from partial studies and superseded analyses. This distinction is used sparingly in the main text and more explicitly in the appendix, where reproduction commands and validation gates are listed.

4. Data Model and Feature Construction

4.1. Canonical Market State

The internal data model is built around a canonical market-state record. Each row is indexed by instrument, timestamp or event index, fold, horizon and data source. The feature side stores book-derived quantities available at the decision point: mid-price, spread, top-of-book depth, multi-level imbalance, microprice displacement, realised-volatility proxy, event intensity where available and source-specific replay diagnostics. The label side stores future-horizon outcomes: direction, return bucket, volatility regime, spread-state transition and adverse-selection proxy where the source supports it.

This schema is the bridge between heterogeneous data and diagnostic sources. FI-2010 contributes normalised snapshot matrices. The synthetic extension contributes explicit event streams, known regimes and generated book states. The aggregate L2 extension contributes snapshot-plus-diff reconstruction state. Because all downstream reports consume the same conceptual schema, model comparisons and diagnostics remain aligned even when the underlying data source changes.

4.2. Feature Groups

Feature groups are registered explicitly rather than treated as anonymous matrix columns. This design supports reproducible ablations and clearer interpretation. Spread and price-level features describe the geometry of the book. Size-level and imbalance features describe visible liquidity. Volatility and event-intensity features describe local activity. The snapshot order-flow proxy captures signed changes inferred from consecutive FI-2010 snapshots. In the synthetic extension, event-level OFI, cancellation imbalance and trade imbalance are computed directly from the event stream.

4.3. Data Quality Checks

The project treats data quality as part of the modelling problem. Replay paths check crossed books, negative depth, missing sequence updates and stale reconstruction state. Benchmark paths check row counts, fold availability, horizon compatibility, preprocessing scope and report freshness. Report builders validate that expected summaries exist before producing claim-bearing tables. These checks reduce the distance between code execution and paper results: a result is not merely computed, it is admitted into the provenance ledger only after its surrounding reports pass structural inspection.

This is particularly important for microstructure data because small structural defects can create large predictive reports. A single leaked transformation, a misaligned horizon or a dropped update sequence can make an experiment look stronger than it is. CHRONOSLOB therefore embeds validation in the data path, the experiment path and the reporting path.

Table 1. Data and diagnostic sources and their research roles.

Source	Primary use	Signal extracted	Scope
FI-2010 snapshots	Public LOB forecasting benchmark	Mid-price direction, calibration, feature stability	Snapshot benchmark results across folds and horizons
Synthetic event stream	Controlled event-level stress tests	Event OFI, cancellation imbalance, trade imbalance, regime labels	Controlled simulator results with known generative regimes
Aggregate L2 replay	Local book reconstruction validation	Update continuity, level-update replay, book invariants	Aggregated depth engineering results
Execution diagnostic suite	Forecasting-to-signal diagnostic bridge	Active fraction, turnover proxy, cost proxy, latency and adverse-selection sensitivity	Offline proxy diagnostics over reports and manifests
Public summary	Traceability and release audit	Report status, generation path, reports	Provenance layer for empirical statements

Table 2. Representative feature groups used across the validation stack.

Group	Role
Spread and price levels	Book geometry and short-horizon friction state
Size levels and depth	Visible liquidity across price levels
Top-book imbalance	Immediate bid-ask pressure
Multi-level imbalance	Depth asymmetry beyond the best quotes
Microprice offset	Liquidity-weighted displacement from mid-price
Snapshot order-flow proxy	Signed state changes inferred from FI-2010 snapshots
Event OFI and cancellations	Event-stream pressure in synthetic replay
Volatility and intensity	Local activity and regime context

5. Leakage-Safe Forecasting Protocol

5.1. Labels

Let x_t denote the feature vector observable at snapshot or event index t, and let m_t denote the mid-price. For horizon h, the future return proxy is

$$r_{t,h} = \frac{m_{t+h} - m_t}{m_t}. \tag{1}$$

The directional target is obtained by thresholding the future return:

$$y_{t,h} = \begin{cases} \text{up}, & r_{t,h} > \tau_h, \\ \text{down}, & r_{t,h} < -\tau_h, \\ \text{stationary}, & |r_{t,h}| \leq \tau_h. \end{cases} \tag{2}$$

This definition separates observation time from label time. Features at t are computed from past and current information; labels are computed from future horizons after t. The threshold τ_h defines the stationary region and controls the class balance at each horizon.

For selective prediction, a model outputs class probabilities $p_t(k)$ and confidence

$$c_t = \max_k p_t(k). \tag{3}$$

At threshold γ, the active set is

$$\mathcal{A}_\gamma = \{t : c_t \geq \gamma\}, \tag{4}$$

with active fraction

$$A(\gamma) = \frac{|\mathcal{A}_\gamma|}{N}. \tag{5}$$

All confidence-filtered metrics are interpreted jointly with $A(\gamma)$, because a highly accurate subset can become empirically uninformative if the coverage is too small.

5.2. Temporal Splits and Train-Only Preprocessing

The core protocol is chronological. Validation data are carved from training data, and test data are not used to fit preprocessing transformations, select checkpoints, tune thresholds or calibrate models. This is the correct discipline for financial time series: the future cannot influence the representation of the past. Matched comparisons require identical fold, horizon, seed, lookback, feature construction, preprocessing and architecture, with the objective as the controlled difference.

This rule is especially important for SSL comparisons. Self-supervised learning can change representation quality, optimisation dynamics and calibration simultaneously. CHRONOSLOB therefore evaluates SSL only through matched cells. The first-generation SSL study, SSL-v2 study and validation-selected benchmark are not blended into one universal ranking; each study answers a distinct question under its own benchmark scope.

5.3. Metrics

The primary metrics are accuracy, macro-F1, Matthews correlation coefficient, expected calibration error and Brier score. Macro-F1 is central because FI-2010 class balance varies across horizons and folds. MCC measures multiclass association in a way that is less sensitive to trivial majority-class prediction than accuracy alone (Matthews, 1975). ECE and Brier score measure probabilistic quality, which matters once the model is used as a selective predictor. For confidence bins $B_1, \dots, B_M$, the expected calibration error is

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \qquad (6)$$

and the multiclass Brier score is

$$\mathrm{Brier} = \frac{1}{N} \sum_{t=1}^{N} \sum_{k=1}^{K} \left(p_t(k) - \mathbb{I}[y_t = k]\right)^2. \qquad (7)$$

These definitions make probability quality observable before any selective or execution-aware diagnostic is applied.

The execution-aware layer introduces diagnostic quantities derived from the active set. Let $s_t \in \{-1, 0, 1\}$ denote the model-implied direction, with zero representing abstention. A simple turnover proxy can be written as

$$T = \sum_{t=2}^{N} |s_t - s_{t-1}|, \qquad (8)$$

and a latency diagnostic evaluates predictions after a row-step delay ℓ, replacing s_t by $s_{t-\ell}$. A configured-cost proxy subtracts stylised spread and fee terms from the directional reward proxy. These quantities are deliberately auditable: they are simple enough to trace back to predictions and rich enough to expose the forecasting-versus-signal-quality gap.

6. Model Families

6.1. Classical Baselines

The classical benchmark contains majority class, logistic regression, ridge, elastic-net, random forest and gradient boosting models. These models are not cosmetic baselines. LOB snapshots contain strong low-dimensional structure, and classical learners can exploit imbalance, spread and level-derived features. A neural model that only exceeds a majority baseline is not persuasive; a neural benchmark becomes meaningful only when the classical reference set is explicit.

The classical results remain useful throughout the paper. They anchor FI-2010 forecasting performance, supplies stable reference metrics and provides a contrast to neural experiments whose performance depends more heavily on lookback, objective and training schedule.

6.2. Matrix Transformer

The matrix transformer treats a lookback window of FI-2010 matrices as a numerical sequence. Each row is projected into a compact latent space, augmented with a learned positional embedding and passed through a transformer encoder. The implementation uses a linear input projection, model dimension 16, two attention heads, one encoder layer, feed-forward width 32, GELU activation, mean pooling over the lookback window and a linear classification head. The benchmark sweeps lookbacks 20, 50 and 100; the maximum sequence length is set to the corresponding lookback.

A simplified description is

$$h_{t-L+1:t} = \mathrm{Transformer}(PX_{t-L+1:t} + E_{1:L}), \qquad (9)$$

$$z_t = \frac{1}{L} \sum_{i=1}^{L} h_{t-L+i}, \qquad (10)$$

$$\hat{p}_t = \mathrm{softmax}(Wz_t + b), \qquad (11)$$

where $X_{t-L+1:t}$ is the lookback window, P is the input projection, E is the learned positional embedding and z_t is the pooled sequence representation. The architecture is deliberately aligned with snapshot matrices rather than raw order events, which makes it a clean reference model for FI-2010.

6.3. DeepLOB-Style Model

The DeepLOB-style model provides a convolutional–recurrent reference family inspired by the canonical DeepLOB design. It receives tensors of shape $[\mathrm{batch}, \mathrm{lookback}, \mathrm{features}]$, applies two one-dimensional convolutional blocks over the feature channels, uses batch normalisation and ReLU activations, then feeds the sequence representation into a one-layer LSTM. The implementation uses 16 convolution channels, kernel size 3, LSTM hidden size 32, dropout 0.1 and a linear classifier. Its role is to supply a compact, auditable CNN–LSTM baseline with the expected LOB inductive bias: local cross-sectional structure followed by temporal aggregation.

6.4. Self-Supervised Objectives

The first-generation SSL objectives are masked reconstruction and next-field prediction. Masked reconstruction asks the model to recover hidden input fields. Next-field prediction asks it to forecast subsequent matrix fields. The results from these objectives are scientifically valuable because they demonstrate that generic pretext learning does not automatically improve downstream LOB forecasting.

SSL-v2 changes the pretext objective. Instead of treating reconstruction as the central task, it uses a market-state multitask objective designed around quantities more closely aligned with forecasting and calibration. A generic form is

$$\mathcal{L}_{\mathrm{SSLv2}} = \lambda_1 \mathcal{L}_{\mathrm{state}} + \lambda_2 \mathcal{L}_{\mathrm{vol}} + \lambda_3 \mathcal{L}_{\mathrm{spread}} + \lambda_4 \mathcal{L}_{\mathrm{recon}}, \qquad (12)$$

where the terms represent market-state classification, volatility regime prediction, spread-state prediction and reconstruction-style regularisation. Fine-tuning then evaluates whether the representation improves mid-price forecasting and probability quality under matched conditions.

7. Training and Selection Protocol

7.1. Optimisation Discipline

Neural experiments use validation-only model selection. Training data drive gradient updates, validation data determine early stopping and checkpoint selection, and test data remain sealed until the final evaluation. The validation-selected benchmark uses a maximum of 25 epochs, patience 5, batch size 1024, best-checkpoint restoration and a single official held-out test pass per run. This converts the neural comparison from a fixed-duration smoke test into a controlled model-family study.

The design is deliberately factorial. Folds measure chronological variation, seeds measure optimisation variation, lookbacks measure state-history sensitivity and horizons measure target difficulty. Reporting means and standard deviations across the grid is therefore more informative than selecting a single best configuration.

7.2. Matched-Cell Principle

A central rule in CHRONOSLOB is the matched-cell principle. A comparison is admissible only when all non-objective dimensions are aligned. For two objectives a and b, a matched cell has the same dataset, fold, horizon, seed, lookback, preprocessing path, architecture and evaluation metric. The comparison delta is

$$\Delta_m(a, b; c) = m(a, c) - m(b, c), \tag{13}$$

where m is a metric and c is the shared comparison cell. Aggregate deltas are then computed over the matched cell set,

$$\overline{\Delta}_m = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \Delta_m(a, b; c). \tag{14}$$

This is the basis for the SSL-v2 table: the result is not a loose comparison between separately tuned experiments, but a mean over 30 aligned cells.

7.3. Calibration and Selective Reliability

Probability quality is treated as a first-class modelling output. ECE partitions predictions into confidence bins and compares average confidence to empirical accuracy, while Brier score evaluates squared probability error across classes. These metrics matter because execution-aware diagnostics use confidence as a gating variable. A model with strong raw classification metrics but poor calibration can produce unstable high-confidence subsets. Conversely, a model with moderate raw predictive gains and better probability behaviour can be more compelling once selective prediction is introduced.

Table 3. Evaluation layers used by CHRONOSLOB.

Layer	Principal question
Forecasting	Does the model predict future mid-price direction?
Calibration	Are probability forecasts numerically reliable?
Selective prediction	What remains after confidence filtering?
Activity	How much of the stream is active?
Cost and latency	How sensitive is the signal proxy?
Results ledger	Which report supports the statement?

Table 4. Validation-selected neural benchmark scope.

Dimension	Values
Folds	1–5
Seeds	0–2
Lookbacks	$20, 50, 100$
Horizons	$10, 50$
Model families	2
Runs per family	90
Total cells	$180/180$
Failures	0

7.4. Reporting Standard

The reporting standard is intentionally strict. Tables report means across benchmark scopes, not isolated maxima. Figures expose active fraction alongside filtered metrics. Text describes feature effects as stability analysis, not causal proof. Replay extensions are reported as reconstruction and diagnostics infrastructure. This produces a paper that is ambitious because it is disciplined: strong empirical results are presented clearly, and the underlying protocol is visible enough for the reader to audit.

8. Validation-Selected Neural Benchmark

8.1. Benchmark Design

The v0.3.0 neural benchmark completes the main empirical comparison in the project. It covers 180 planned training cells with zero failures. The factorial design spans folds 1–5, seeds 0–2, lookbacks 20/50/100, horizons 10/50 and two model families: matrix transformer and DeepLOB-style. Each model family therefore contributes 90 runs. Validation-only early stopping and best-checkpoint restoration make the comparison materially stronger than a fixed one-epoch grid.

The benchmark is designed to test robustness across three

Table 5. Validation-selected neural benchmark across 90 runs per model family. Values are mean $\pm$ standard deviation across folds, seeds, lookbacks and horizons.

Model family	Macro-F1	MCC
Matrix transformer	0.6013 ± 0.1950	0.4294 ± 0.2727
DeepLOB-style	0.5133 ± 0.0670	0.3356 ± 0.0883

sources of variation. Folds test chronological market segments. Seeds test optimisation variation. Lookbacks test how much recent history each architecture uses. This design is more informative than a single best run because it measures the stability of a model family under repeated conditions.

8.2. Aggregate Results

Table 5 reports aggregate benchmark means. The matrix transformer has higher mean macro-F1 and MCC in the benchmark, while the DeepLOB-style family provides an important convolutional-recurrent reference. The larger transformer dispersion indicates higher configuration sensitivity rather than uniformly dominant performance across all cells. The comparison is deliberately reported at family level rather than as a single cherry-picked configuration.

The result gives the paper its strongest neural modelling result. The matrix transformer benefits from a sequence representation aligned with the snapshot window and attention-based temporal aggregation. The DeepLOB-style family remains competitive but lower on the mean metrics. The more important point is methodological: both models are evaluated under the same grid, so the comparison is not confounded by fold, seed, horizon or lookback mismatch.

8.3. Lookback and Confidence Behaviour

The lookback dimension is central because market microstructure contains both immediate pressure and decaying contextual information. Shorter windows can emphasise the most recent book imbalance; longer windows can capture broader state but may also dilute local signal. The benchmark shows that lookback is not a trivial scaling knob. Performance concentrates in configurations where the history length is aligned with the horizon and model family.

Confidence filtering adds another view. Higher thresholds improve selected-sample metrics but reduce active fraction. This is precisely why CHRONOSLOB reports confidence-filtered performance with coverage. A model family can be strong in raw predictive metrics yet operationally selective at high confidence. Conversely, a model with moderate raw scores may provide a usable subset if its confidence is better aligned with correctness.

Table 6. SSL-v2 matched aggregate deltas across 30 FI-2010 comparison cells. Positive macro-F1 and MCC deltas are favourable; negative ECE and Brier deltas are favourable.

Metric	Mean delta
Macro-F1	$+0.0112$
MCC	$+0.0259$
ECE	-0.0030
Brier score	-0.0235

9. SSL-v2 Benchmark

9.1. Motivation

The first-generation SSL study is a useful negative result: generic masked reconstruction and next-field objectives do not provide broad downstream improvement in the matched grid. SSL-v2 is designed as a more market-aware alternative. Instead of reconstructing arbitrary fields, the objective forces representations to organise around market-state quantities that are closer to downstream forecasting and calibration.

The SSL-v2 cluster result covers folds 1–5, horizons 10/50, seeds 0–2 and lookback 50. Across 30 matched comparison cells, supervised and SSL-v2 runs are compared under identical fold, horizon, seed and lookback. This design isolates the contribution of the objective.

9.2. Results

Table 6 shows that SSL-v2 improves the mean matched comparison across both predictive and probabilistic metrics. The deltas are modest but coherent: macro-F1 and MCC increase, while ECE and Brier score decrease. In a noisy microstructure benchmark, this is a meaningful empirical profile because it combines classification improvement with better probability behaviour.

The heterogeneous cell-level pattern remains important for interpretation. In market microstructure, improvements rarely distribute uniformly across folds, seeds and horizons. CHRONOSLOB therefore treats SSL-v2 as a scoped representation result rather than as an architecture slogan. Its value is that it moves the results from generic SSL failure toward a market-state objective with measurable gains under a matched multi-seed scope.

10. Feature-Stability Analysis

10.1. Ablation Design

Feature-stability analysis asks whether removing a feature group changes predictive performance under controlled conditions. The expanded FI-2010 ablation results contain 2,580 completed rows: logistic and ridge models across folds 1–5, horizons 10/20/50, seeds 0–2, 12 registry groups

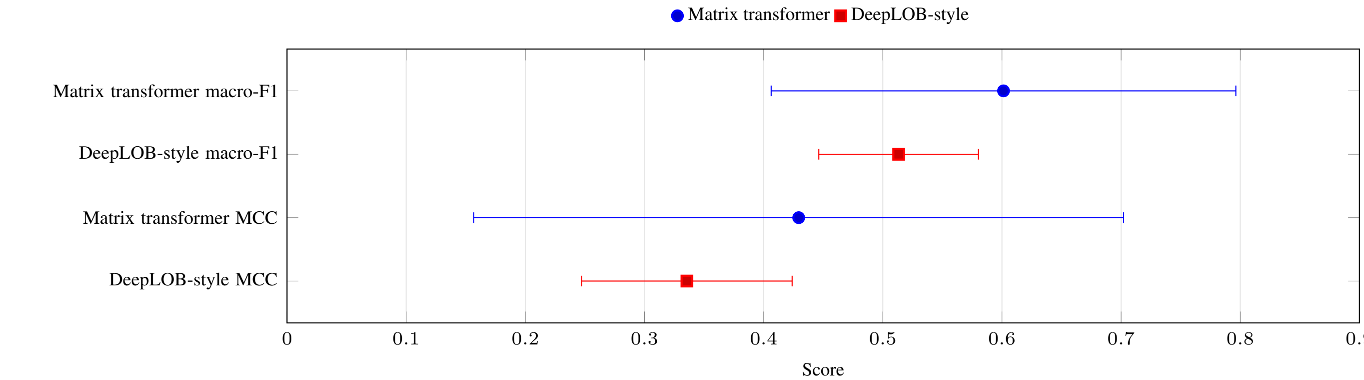

Figure 2. Validation-selected neural benchmark with one-standard-deviation intervals across the 180-cell grid. The matrix transformer has stronger means, while the intervals expose substantial cross-fold, cross-seed, cross-lookback and cross-horizon dispersion.

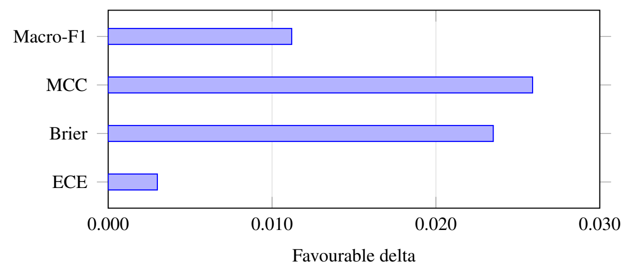

Figure 3. SSL-v2 aggregate deltas across 30 matched comparison cells. ECE and Brier are sign-adjusted so that all bars point in the favourable direction.

Table 7. Feature-stability summary for removing the snapshot-derived order-flow proxy. Negative deltas indicate degradation after removal.

Scope	Rows degraded	Mean Δ F1
Horizon-10 logistic/ridge	30/30	−0.1957
Horizon-20/50 matched scope	65/65	−0.1032
Gradient-boosting slice	10/10	−0.3369

and six ablation modes, plus a small gradient-boosting nonlinear slice. The purpose is not causal inference. It is stability measurement: if a feature group consistently degrades performance when removed, the group is an important component of the predictive representation.

10.2. Snapshot Order-Flow Proxy

The strongest feature effect is the `snapshot_order_flow_proxy`. Removing this group degrades macro-F1 in every matched horizon-10 logistic/ridge row, every broader horizon-20/50 matched removal row and every row in the small nonlinear slice. Table 7 summarises the effect.

The terminology is deliberately exact. In FI-2010 this is a snapshot-derived proxy, not event-level OFI. Its empirical importance is still substantial: it captures changes in observed book states that are strongly associated with future mid-price direction. The synthetic event-level extension separately validates event-level OFI-style features when the underlying event stream is known.

Table 8. Synthetic event-level replay quality.

Diagnostic	Value
Events	21,010
Regimes	7
Snapshots	4,202
Crossed books	0
Negative-depth states	0
Sequence gaps	0

11. Replay Extensions

11.1. Synthetic Event-Level LOB

The synthetic extension expands CHRONOSLOB beyond static snapshot matrices. It generates 21,010 events across seven regimes: stable liquid, buy pressure, sell pressure, high volatility, low liquidity, wide spread and cancellation shock. The replay engine converts the event stream into 4,202 snapshots with zero crossed books, zero negative-depth states and zero sequence gaps. This validates the pipeline from events to local book states, features, labels and diagnostics.

The synthetic benchmark also produces a chronological model comparison: majority macro-F1 approximately 0.21, logistic regression 0.34, ridge 0.35 and gradient boosting 0.48. These results show that the simulated regimes produce learnable structure while retaining enough complexity to stress the feature and label path.

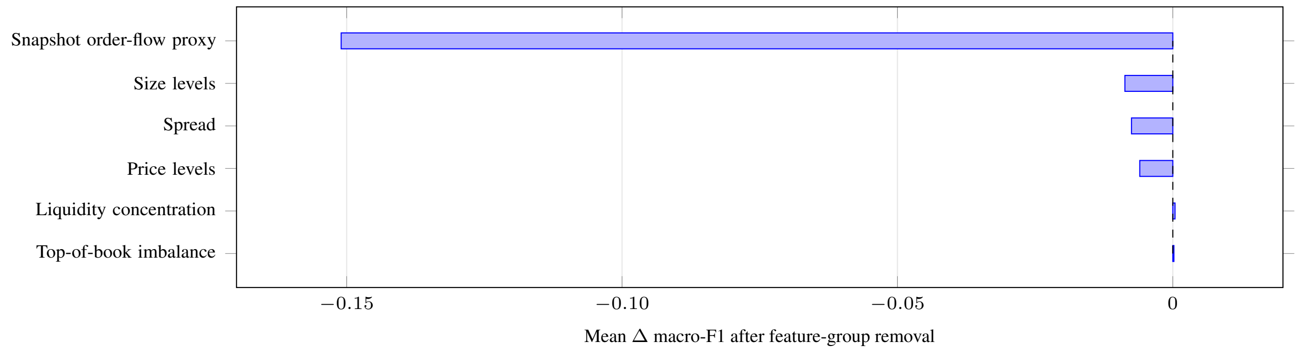

Figure 4. Ranked remove-one-group stability effects. Negative values indicate degradation after removal; the snapshot-derived order-flow proxy is the dominant FI-2010 feature group in this analysis.

11.2. Aggregate L2 Replay

The aggregate L2 replay path implements snapshot-plus-diff reconstruction, update-continuity checks, book-invariant validation and summary-level reporting. It provides an engineering bridge between offline benchmark matrices and exchange-style level updates. The system loads an initial snapshot, buffers and applies diff-depth updates, checks update identifiers and emits replay summaries. This component strengthens the project because it validates local-book mechanics independently of the FI-2010 benchmark.

The distinction between aggregate level updates and event-level order streams is reflected in the design. Aggregate L2 replay validates price-level reconstruction and continuity, while synthetic replay validates event-level feature calculation under controlled events. Together they form a broader market-data substrate: static snapshots for benchmark forecasting, generated events for controlled event diagnostics and aggregate diffs for local-book reconstruction.

12. Execution-Aware Diagnostics

12.1. Forecasting Versus Signal Quality

The execution diagnostic suite is the core diagnostic result of CHRONOSLOB. It joins forecast summaries to confidence thresholds, active fraction, turnover proxy, configured-cost sensitivity, row-step latency and adverse-selection proxy. The diagnostic asks a sharper question than standard forecasting tables: how does the interpretation of the same model change once its predictions are filtered and stressed?

Table 9 captures the paper's central empirical motif. At low thresholds, activity is broad and the configured-cost proxy is strongly negative. At high thresholds, the proxy improves as the selected set becomes extremely small. This is exactly why selective prediction cannot be summarised by a single selected-sample score. The active fraction is part of the result.

Table 9. Confidence-threshold trade-off from the execution diagnostic suite.

Threshold	Active fraction	Cost proxy
0.33	0.544020	−6159.015
0.50	0.216300	−2011.700
0.70	0.017206	−146.894
0.85	0.002096	−17.968
0.95	0.001290	11.750

12.2. Metric-to-Proxy Gap

Table 10 shows why a forecast table is not a complete market-microstructure evaluation. The supervised and masked-reconstruction rows have similar macro-F1, yet their active fraction, cost proxy and adverse-selection proxy differ. The next-field row has lower macro-F1 but a different activity profile. This is not a contradiction; it is the distinction between prediction and signal diagnostics made measurable.

12.3. Cost and Latency Sensitivity

Cost sensitivity and latency sensitivity provide disciplined stress tests. Increasing configured fees and spread multipliers worsens the cost-adjusted proxy. Increasing row-step latency degrades signal quality relative to the no-lag setting, with the largest degradation around horizon 50 in the execution-v3 analysis. These results align with the microstructure intuition that short-horizon forecasts decay as market state updates. The diagnostic value is that the decay is quantified and attached to reports and manifests.

The adverse-selection proxy adds another layer. In the analysis, lower-confidence predictions have a higher adverse-selection proxy than selected high-confidence buckets, while the selected buckets also have lower coverage. This is the same trade-off from another angle: confidence can improve diagnostics, but only after the active fraction is accounted for.

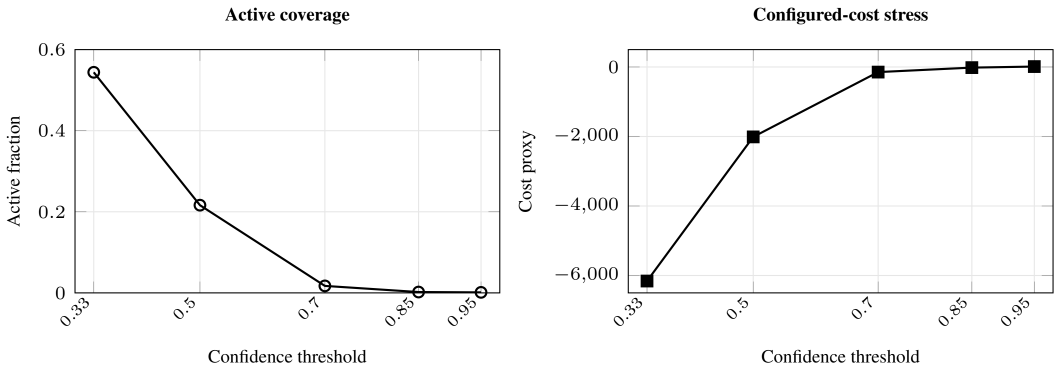

Figure 5. Confidence filtering trade-off. Higher thresholds sharply reduce active coverage while moving the configured-cost proxy towards zero; the high-threshold regime must therefore be interpreted jointly with its selected-sample size.

Table 10. Representative horizon-50 metric-to-proxy rows. Similar forecasting metrics can lead to different confidence, cost, latency and adverse-selection profiles.

Objective	Macro-F1	ECE	Active @0.70	Cost @0.70	Latency degradation	Adv. proxy
Supervised	0.4180	0.0733	0.0152	−26.9	−2476.7	0.3297
Masked reconstruction	0.4148	0.0854	0.0330	−101.3	−2236.5	0.2019
Next field	0.3823	0.0846	0.0543	−219.8	−1974.1	0.2770

13. Provenance and Reproducibility

13.1. Provenance

The v0.3.0 benchmark is connected to generated summaries, configuration snapshots, validation reports and benchmark manifests. Cluster-execution details are recorded in the project provenance files and summarised in the appendix; the main paper focuses on the experimental design and results. This separation keeps the manuscript technical without exposing operational detail in the central narrative.

The repository is structured so that a reader can trace results from paper tables to generated reports, from reports to CLI entry points and from entry points to source modules. The paper is therefore a synthesis of an executable research system rather than a detached write-up.

13.2. Reproducibility Commands

The principal reporting commands are listed in Appendix A. They cover SSL analysis, execution diagnostics, feature ablations, synthetic benchmarking, aggregate L2 replay, report generation, public summaries and strict project audit. The validation gates combine tests, linting, type checking, project diagnostics, release-readiness inspection and whitespace validation.

14. Discussion

CHRONOSLOB demonstrates a mature empirical pattern for market-microstructure research. Predictive modelling is necessary, but insufficient as a final evaluation layer. The platform first constructs temporally valid benchmark data, then trains matched models, then evaluates prediction and calibration, then applies selective prediction, and finally stresses forecasts through activity, configured costs, latency and adverse-selection diagnostics.

The 180-cell validation-selected benchmark provides the strongest supervised neural comparison in the paper. The matrix transformer has higher mean macro-F1 and MCC than the DeepLOB-style family across the full grid, while its larger standard deviations reveal materially higher configuration sensitivity. SSL-v2 contributes a separate representation-learning result: market-state multitask pretraining improves matched mean deltas across 30 cells. Feature ablations show that snapshot-derived order-flow information is central to FI-2010 feature stability. Replay extensions and execution diagnostics then widen the analysis beyond static forecast tables.

The main research message is that evaluation design is part of the modelling contribution. A market-microstructure paper becomes substantially more informative when it reports what changes under calibration, coverage and friction stress rather than presenting a single predictive metric as the endpoint.

15. Limitations

FI-2010 is a snapshot benchmark, so it supports leakage-safe forecasting and feature-stability analysis but does not expose order identifiers, queue position, realised fills or venue-specific matching rules. The snapshot-derived order-flow proxy is therefore a powerful benchmark feature, not a substitute for event-level order-flow imbalance. The synthetic extension validates event-level machinery under controlled regimes, while aggregate L2 replay validates snapshot-plus-diff reconstruction for level updates. Execution diagnostics are stylised proxy tests over model outputs; richer realised-fill analysis would require venue data with order identifiers, precise timestamps, fees, queue priority and market-impact observability.

These boundaries sharpen rather than weaken the result. They define the next empirical frontier: higher-resolution market data can enter the existing data and replay layers, reuse the same feature and label interfaces, train through the same model layer and report through the same diagnostic protocol.

16. Conclusion

CHRONOSLOB presents a rigorous research platform for market-microstructure forecasting. It combines leakage-safe FI-2010 experiments, supervised and self-supervised neural objectives, a completed 180-cell validation-selected benchmark, feature-stability analysis, event-level replay extensions, aggregate L2 reconstruction and execution-aware diagnostics. The results show strong matrix-transformer performance in the validation-selected benchmark, positive SSL-v2 mean deltas under matched multi-seed comparison and a clear divergence between predictive metrics and signal-quality diagnostics.

The methodological conclusion is direct. Market-microstructure systems should report predictive metrics, calibration, confidence-filtered coverage and execution-aware diagnostics together. CHRONOSLOB implements that principle as code, reports and paper: a forecasting platform whose empirical strength comes from the coherence of its full validation protocol.

Reproducibility Statement

The repository contains source code, experiment runners, generated reports, public summaries and quality gates. Aggregate summaries, manifests and reports are versioned for inspection; large intermediate prediction files and checkpoints are excluded where they are not required for the manuscript. The commands in Appendix A connect the paper to executable modules and generated reports.

References

Brier, G. W. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

Cont, R., Kukanov, A., and Stoikov, S. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

Dawid, A. P. The well-calibrated bayesian. *Journal of the American Statistical Association*, 77(379):605–610, 1982.

Gould, M. D., Porter, M. A., Williams, S., McDonald, M., Fenn, D. J., and Howison, S. D. Limit order books. *Quantitative Finance*, 13(11):1709–1742, 2013.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.

Lopez de Prado, M. *Advances in Financial Machine Learning*. Wiley, 2018.

Matthews, B. W. Comparison of the predicted and observed secondary structure of t4 phage lysozyme. *Biochimica et Biophysica Acta*, 405(2):442–451, 1975.

Ntakaris, A., Magris, M., Kanniainen, J., Gabbouj, M., and Iosifidis, A. Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*, 37(8):852–866, 2018.

Sirignano, J. and Cont, R. Universal features of price formation in financial markets: Perspectives from deep learning. *Quantitative Finance*, 19(9):1449–1459, 2019.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Zhang, Z., Zohren, S., and Roberts, S. Deeplob: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.

A. Reproducibility Surface

A.1. Command Map

The repository exposes the paper's empirical layers through explicit CLI entry points. The SSL commands rebuild the first-generation and SSL-v2 analyses; the execution commands rebuild the forecasting-versus-signal-quality reports; the replay commands rebuild the synthetic and aggregate L2 paths; the reporting commands rebuild the public summaries.

Table 11. Core reproduction commands.

Command	Purpose
`analyse-fi2010-ssl-results`	SSL-v1 analysis
`analyse-fi2010-ssl-v2-results`	SSL-v2 matched comparison
`analyse-fi2010-execution-v3`	Execution-v3 diagnostics
`build-execution-centrepiece`	Signal-quality reports
`analyse-fi2010-feature-ablations`	Feature-stability analysis
`run-synthetic-lob-benchmark`	Synthetic event benchmark
`replay-binance-l2-sample`	Aggregate L2 replay
`build-final-empirical-report`	Generated report
`build-evidence-pack`	Public summary
`run-project-audit--strict`	Project audit

Table 12. Validation commands.

Command	Purpose
`python-mpytest-q`	Test suite
`python-mruffcheck.`	Linting
`python-mmypychronoslob`	Static typing
`python-mchronoslob.clidoctor`	Project diagnostics
`inspect-release-readiness`	Release inspection
`run-project-audit--strict`	Strict project audit
`gitdiff--check`	Whitespace validation

A.2. Validation Gates

Release-readiness combines tests, linting, static typing, project diagnostics, release inspection, strict audit and whitespace validation. The gates are intentionally heterogeneous: they check numerical code, written reports, provenance paths and repository hygiene.

A.3. Compact Results Tables

B. Implementation Notes

The codebase is organised into modules for data handling, feature construction, labels, models, training, analysis, replay, reporting and audit. This organisation makes the paper inspectable: generated results link to commands, commands resolve to source modules and reports record the relevant scope. The written paper and the executable repository therefore share the same structure.

The same architecture also keeps future extensions natural. A richer event-level source can enter through the data and replay layers, reuse feature and label interfaces, train through the existing model layer and report through the same diagnostic and provenance path. The project therefore functions as a reusable research instrument rather than a one-off collection of experiments.

Table 13. Synthetic chronological benchmark.

Model	Test macro-F1
Majority	0.21
Logistic regression	0.34
Ridge	0.35
Gradient boosting	0.48

Table 14. Execution diagnostic active fraction.

Threshold	Active fraction
0.33	0.544020
0.70	0.017206
0.85	0.002096
0.95	0.001290

Table 15. Notation used in the forecasting and diagnostic layers.

Symbol	Meaning
m_t	Mid-price at snapshot or event index t
h	Forecast horizon in rows or events
$r_{t,h}$	Future return proxy from t to $t+h$
$p_t(k)$	Predicted probability of class k
c_t	Maximum predicted confidence
γ	Confidence threshold for active predictions
$A(\gamma)$	Active fraction after confidence filtering
s_t	Directional signal after abstention logic
ℓ	Row-step latency offset

C. Notation Summary

Table 15 summarises the notation used in the forecasting and diagnostic layers.

D. Paper-to-Repository Mapping

The paper follows the repository's empirical structure. FI-2010 forecasting corresponds to the benchmark and neural-training experiment directories; SSL-v2 corresponds to the market-state multitask training and analysis modules; feature stability corresponds to the ablation reports; replay results correspond to the synthetic and aggregate L2 packages; execution-aware analysis corresponds to the execution diagnostic suite reports; and the provenance ledger corresponds to the generated results pack and strict audit command. This mapping makes the manuscript directly inspectable from the codebase.